

Wi-Fi Security Protocol

Module 31

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Hello friends I am Professor yogesh Chaba from Department of computer science and engineering Guru jambheshwar University of Science and Technology Hisar harana today in this lecture we will study about Wi-Fi technology first of all please see the contents which I shall cover after covering brief introduction about Wi-Fi technology I shall take up its layered protocol architecture which will be followed by Wi-Fi standards in last I shall cover WiFi frame format a wireless local area network W Lan is a data Communication System implemented as an extension or as an alternative to a wired LAN WiFi is a technology of Wireless local area network based on the i e 802.11 series of Wireless standards issued by the Institute of electrical and

electronic Engineers i e an international Professional Organization for electrical and electronic Engineers the i e does not test equipment for 2 minutes, 1 second compliance with their standards the nonprofit Wi-Fi Alliance was formed in 1999 to establish and enforce standards for interoperability and backward compatibility and to promote Wireless local area Network Technology the Wi-Fi Alliance enforces the use of the Wi-Fi brand to Technologies based on the i e 802.11 standard manufacturers with membership in the Wi-Fi Alliance whose products are manufactured as per I E 802.11 standard gain the right to mark their products with the Wii logo specifically the certification process requires compliance to the i e 802.11 radio standards the

WPA and WPA2 security standards and the EAP authentication standards any manufacturer who manufactures product as per the standard can put a Wi-Fi logo on their product there are several Technologies to choose from while selecting a wireless LAN solution each with advantages and limitations most Wireless lens use spread Spectrum a whiteband radio frequency technique in which the signal is spread out over the available bandwidth there are two types of spread Spectrum first is frequency hoping and second direct sequence frequency hoping spread Spectrum FHS uses a narrow band carrier that changes frequency in a pattern known only to the transmitter and receiver on the other hand direct sequence spread Spectrum DS generates a red redundant bit pattern

for each bit to be transmitted and requires more bandwidth for implementation this bit pattern called a chip or chipping code is used by the receiver to recover the original signal even if one or more bits in the chip are damaged during transmission statistical technique embedded in the radio can recover the original data without the need for retransmission now let us move on to the layered protocol architecture of i.e. 802.11 the i.e. 802.11 standard only covers the physical layer and medium access layer of OSI reference model as shown in figure the physical layer is subdivided into physical layer convergence protocol (PLCP) and the physical medium dependent sublayer (PMD) the PLCP sublayer provides a carrier

sense signal called Clear Channel Assessment (CCA) and provides a common physical service access point (SAP) independent of the transmission technology the second sublayer (PMD) handles modulations and encoding or decoding of signals the MAC management support medium Access Control Association and re Association of a station to an access point and roaming between different access points it also controls authentication mechanism encryption synchronization of a station with regard to an access point and power management to save battery power further 802.11 uses frequency hopping spread Spectrum (FHSS) direct sequence spread Spectrum (DSSS) orthogonal frequency Division multiplexing (OFDM) and high rate direct sequence

spread Spectrum (HR DSSS) modulation techniques at physical layer frequency hopping spread Spectrum (FHSS) uses 79 channels of 1 MHz each and a pseudo random number generator is used to produce the sequence of frequencies hopped to this method provides a fair way to allocate Spectrum in the unregulated ISM band direct sequence spread Spectrum (DSSS) technique supports 8 minutes, 1 second communication speed of 1 or 2 Mbps it basically uses a technique similar to code division multiple access (CDMA) next modulation technique used in physical layer is orthogonal frequency division multiplexing (OFDM) which delivers 54 Mbps and above speed in the wider 5 GHz ISM band next technique is high rate direct sequence spread Spectrum which offers speed of 11

and 22 Mbps in 2.4 GHz bandwidth in MAC layer i.e. 802.11 uses two modes of operations the first called distributed coordination function (DCF) and second Point coordination function (PCF) DCF does not use any kind of central control it uses a protocol called carrier sense multiple access with collision avoidance (CSMA/CA) in this method both physical Channel sensing and virtual Channel sensing is done in physical Channel sensing when a station wants to transmit it senses the channel if it is Idle it just starts transmitting it does not sense the channel while transmitting but transmits full frame which may get destroyed at the receiver due to interference if a collision occurs the colliding station waits for a random

amount of time using the ethernet binary exponential backoff algorithm and then try again later in Virtual Channel sensing scheme medium access with collision avoidance (MACA) is used in point coordination scheme (PCF) base station is used to control all activity in its cell now let us see different Wireless LAN standard for i.e. 802.11 i.e. 802.11 is a set of medium access

control and physical layer specification for implementing Wireless local area network in 2.4 and 5 GHz frequency band they are created and maintained by the IEEE standard committee IEEE 802.11 the base version of the standard was released in 1997 and had many subsequent amendments the standard and amendments provide the basis for wireless network products using the wifi brand there are

several Wireless LAN standards each suited for a particular environment as shown in figure first standard of IEEE 802.11 Series is 802.11 itself which was released in 1997 it offers 1 to 2 Mbps speed with an indoor range of 20 m and an outdoor range of 100 m it uses FHSS and DSSS as spread Spectrum technique it operates on 2.4 GHz ISM band next standard is 802.11a which was released in 1999 it offers 54 and 8 Mbps speed with an indoor range of 35 m and outdoor range of 120 m it uses OFDM as modulation technique it operates on 5 GHz ISM band since the 2.4 GHz band is heavily used so using the relatively unused 5 GHz band gives 802.11a significant advantage in theory 802.11a signals are absorbed more readily by walls and other solid objects

in their path due to their smaller wavelength and as a result they cannot penetrate now moving on to IEEE 802.11b this standard was released in 1999 it offers 11 and 22 Mbps speed with an indoor range of 35 m and outdoor range of 120 m it uses HR DSSS as spread Spectrum technique it operates on 2.4 GHz ISM band next is IEEE 802.11g standard this standard was released in 2003 it offers 54 Mbps speed with an indoor range of 38 m and outdoor range of 140 m it uses OFDM and DSSS as spread Spectrum technique it operates on 2.4 GHz ISM band IEEE 802.11n released in 2009 is an amendment that improves upon the previous 802.11 standards by adding multiple input multiple output MIMO antennas

802.11n offers speed of 54 Mbps to 600 Mbps using OFDM spread Spectrum technique it operates on both the 2.4 GHz and the lesser used 5 GHz band support for 5 GHz band is optional this standard offers indoor range of 70 m and outdoor range of 250 m IEEE 802.11AC released in 2013 is built on 802.11n it offers speed in the range of 1300 Mbps and operates on 5 GHz ISM band other IEEE 802.11 standards IEEE 802.11D addresses the spread of technology to countries not addressed by IEEE 802.11e focuses on the quality of service associated with wireless transmission of multimedia applications IEEE 802.11f addresses roaming between APs and interoperability between different vendor groups IEEE 802.11h focuses on frequency selection

and power concerns on the 5 GHz band in European countries IEEE 802.11i focuses on enhancing WLAN security and Authentication for 802.11 that include radius servers and IEEE 802.11x IEEE 802.11j focuses on the Japanese equivalent of 802.11h IEEE 802.11ad is an amendment that defines a new physical layer for 802.11 Network to operate in the 60 GHz Spectrum products implementing the 802.11ad standard are being brought to Market under the Wgig brand name the certification program is now being developed by the wifi Alliance instead of Wgig Alliance the peak transmission rate of 802.11ad is 7 Gbps per second IEEE

802.11 AF also referred to as white or super WiFi is an amendment approved in February 2014 that allows W Lan operation in TV

white space Spectrum in the VHF and UHF bands between 54 and 790 mhz it uses cognitive radio technology to transmit on unused TV channels Wireless use electromagnetic waves to communicate information from one point to another without relying on a wired connection radio waves are often referred to as radio carriers because they simply perform the function of delivering energy to a remote receiver the data being transmitted are superimposed on the radio carrier so that they can be extracted accurately at the receiving end this process is generally referred to as carrier modulation i.e. 802.11b and 802.11g standard utilizes the 2.4 GHz Spectrum however 802.11a used the more heavily regulated 5 GHz band i.e.

802.11n utilizes both 2.4 GHz and 5 GHz band these bands are commonly referred to as 2.4 GHz and 5 GHz band in most of the literature each spectrum is subdivided into Channels with a center frequency and bandwidth analog to the way radio and TV broadcast bands are subdivided the 2.4 GHz band is divided into 14 channels spaced 5 mhz apart beginning with Channel 1 which is centered on 2412 MHz the L channels have additional restrictions or are unavailable for use in some regulatory domains figure shows exact frequencies of 14 different channels used in 2.4 GHz ISM band now we will move on to the frame format of IEEE 802.11 standard frame format of this Wireless standard is as shown in figure there are nine fields in frame format of this standard starting from

frame control to checksum frame control field has 11 subfields now let us see each field one by one frame control field which is of two bytes is used for control purpose details of subfield of frame control field will be discussed after discussing main fields second field is duration which tells how long the frame and its acknowledgement will occupy the channel the frame header contains four addresses the source and the destination are obviously needed and the other two are used for the source and destination base stations for intercell traffic the sequence field allows fragments to be numbered of the 16 bits available 12 identify the frame and four identify the fragment the data field contains the

payload up to 2312 bytes followed by the usual checksum of 4 bytes now let us see details of frame control field which has 11 subfields the first of these is protocol version which allows two versions of the protocols to operate at the same time in the same cell then comes the type field which indicates data control or management type of frame and next field indicates subtype the two DS and from DS bits indicates that the frame is going to or coming from the inter cell distribution system for example ethernet system the MF bit means that more fragments will follow the retry bit marks a retransmission of a frame sent earlier the power management pwr bit is used by the base station to put the receiver into sleep state or

take it out of sleep State the more bit indicates that the sender has additional frames for the receiver the wbit specifies that the frame body has been encrypted using the web wired equivalent privacy algorithm and finally the obit tells the receiver that a sequence of frame with this bit on must be processed strictly in order so dear friends today in this lecture we have covered introductory concepts of Wi-Fi technology followed by its layered protocol architecture and i e 802.11 standards in last we have seen frame format of i e 802.11 technology so this was all about WiFi technology thank you